

Evan Guerra

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RESEARCH INTEREST

My research bridges scientific language and molecular representations to create multi-modal language, molecule, and machine learning models for scientific applications, including biomedicine, chemistry, and neuroscience.

EDUCATION

University of Illinois at Urbana-Champaign

Urbana, IL

PhD Information Science

May 2028

- Cumulative GPA: 4.0
- Co-advisors: Professor Catherine Blake, Professor Halil Kilicoglu
- Field Exam: Large Language Models, Information Extraction, and Chemosensory Science

Pennsylvania Western University at Clarion

Clarion, PA

Master of Science Library and Information Science

December 2022

- Cumulative GPA: 3.9
- Capstone Advisor: Dr. Danielle Reed (full member, Monell Chemical Senses Center)
- Capstone: Digitization in a Smell and Taste Centered Information Institution

The Evergreen State College

Olympia, WA

Bachelor of Arts in Computer Science

December 2018

- GPA Equivalent: Excellent (Scale: Unsatisfactory – Excellent)
- Advisor: Professor Richard Weiss
- Area of Emphasis: Database Design
- Awards and Honors: Faculty Honors, College Bound Scholarship (Full-ride scholarship)

AWARDS, FELLOWSHIPS, AND GRANTS

2026 Lucille M. Wert Scholar Award: PhD Study in Molecular Property Prediction

\$1,500

American Chemical Society Division of Chemical Information

2025 DREAM Olfaction Prediction Machine Learning Challenge

1st Leaderboard, 4th Challenge

2023-2027 Graduate College Fellowship

\$75,000

University of Illinois Urbana-Champaign

2022, 2023 AChemS Travel Fellowship: NLP for Sweetness Preference Extraction

\$2,000

Association of Chemoreception Sciences

2021 Plain Language Abstract Competition Winner

Global Consortium for Chemosensory Research

2019 Hatching Initiative Grant: Trauma-Informed Robotics

\$2,000

Free Library of Philadelphia Foundation

PROFESSIONAL EXPERIENCE

**Monell Chemical Senses Center
University of Pennsylvania**

Philadelphia, PA
September 2025 – June 2026

Visiting PhD Student

- Utilized python to replicate and validate machine learning models from the DREAM Olfaction Prediction ML Challenge, contributing to consortium paper on challenge results.
- Developed python code to run models on new out-of-domain data and conducted analyses to obtain predictions of novel bifunctional molecules that change in odor quality across concentration, resulting in plug and play models designed for use by chemists and neuroscientists.
- Attended relevant seminars from Penn AI, Penn Neuroscience, and the Monell Chemical Senses Center on multi-modal AI, AI for Science, and computational neuroscience.
- Presented findings weekly to an interdisciplinary team of neuroscientists, biochemists, food scientists, and computational chemists.

University of Illinois at Urbana – Champaign

Urbana, IL
May 2023 - Present

Doctoral Fellow

- Utilized python and TensorFlow to implement transformer model fine-tuning techniques within natural language processing (NLP) and machine learning pipelines, resulting in a publication in Database (5-year impact factor: 4.8).
- Integrated molecular representations from LLMs and Generative Artificial Intelligence in machine learning ensemble model, resulting in 1st place on leaderboard and 4th place (behind 3-way tie) in final round for DREAM Olfaction Prediction ML Challenge.
- Presented results on olfactory language modeling, NLP applications in chemosensory science, and machine learning models to an interdisciplinary team and wrote proposal 1-pager detailing strategy for project direction.
- Mentored Masters student on summer data curation project, providing guidance on how to conduct research, reach project milestones, and implementation of machine learning pipelines.

Research Assistant, NLP Intern

May - August 2025

- Utilized python and pytorch to fine-tuned pre-trained language models for scientific rigor prediction, contributing to key publication milestones of \$3 million dollar NIH grant.
- Co-authored manuscript on integrative health informatics resources, databases, and data repositories, published in top Nutrition journal special issue on AI applications.
- Led data curation and annotation teams in accountability to strict deadlines, creating a dataset of ~500 expert annotated data for NLP and machine learning applications and prepared unstructured dataset for panel presentation at the American Medical Informatics Association annual symposium.

Monell Chemical Senses Center

Philadelphia, PA
August 2021 – May 2023

Junior Research Technician (Data Analyst)

- Conducted exploratory and longitudinal big data analytics for smell and taste research and presented analysis to interdisciplinary scientific team in fast-paced research environment.
- Visualized experimental outcomes using R and Python for grant proposals (\$10 million) and publications, adhering consistently to tight deadlines.

PUBLICATIONS AND CONFERENCES

Journal Articles

[4] Advancing Whole-Person Health through Informatics: A Narrative Review of Knowledge Resources for Complementary and Integrative Health

Robin R Austin, Marcelo Fiszman, Evan Guerra, Rui Zhang, Halil Kilicoglu.
The Journal of Nutrition 2026

[2] Integrating Deep Learning Architectures for Enhanced Biomedical Relation Extraction: A Pipeline Approach

Maria Janina Sarol, Gibong Hong, Evan Guerra, Halil Kilicoglu.
Database 2024

Peer-Reviewed Conference Presentations

[6] [WCRI 2026] **Automated Assessment of the Trustworthiness of Biomedical Publications: A Protocol for a Scoping Review**

Evan Guerra, Joe D. Menke, Cesar Bernabe, Xiangji Ying, Colby J Vorland, Andrew W Brown, Riaz Qureshi, Evan Mayo-Wilson, Halil Kilicoglu.

World Conference on Research Integrity, 2026

[5] [MSLD 2026] **Scientific Olfactory Information Extraction: Toward a Unified NLP Framework for Chemosensory Knowledge Discovery**

Evan Guerra, Danni Wu, Sameer Komoravolu, Halil Kilicoglu

Midwest Speech and Language Days, 2026

[3] [ACChemS 2025] **Extracting Smell Disorders as Early Indicators of Neurodegenerative Diseases Using Natural Language Processing**

Evan Guerra, Halil Kilicoglu.

Association of Chemoreception Sciences (ACChemS) 2025

[1] [ACChemS 2023] **Cross-Country Comparison of Sweet Preferences in Online Reviews of Food Products**

Evan Guerra, Valentina Parma

Association of Chemoreception Sciences (ACChemS) 2023

Papers Under-Review

Olfactory Dysfunction in Children and Adolescents with COVID-19: A Meta-Analysis

Vicente Ramirez, Evan Guerra, Matthew Haigh, Miracle Igram, Jennifer E. Douglas, Pamela H. Dalton, Danielle R. Reed, Elisa Dal Bo, Valentina Parma, 2026

TALKS

AI for Smell and Taste

National Chemistry Week Seminar, Connecticut State Community College – Tunxis, Fall 2025

Extracting Olfactory Information Using NLP

School of Information Sciences Colloquium, UIUC, Spring 2025

How Important is Language in Smell and Taste Research? The case for chemosensory NLP

Illinois Summer Research Symposium, Summer 2023

Database Best Practices

Monell Chemical Senses Center Responsible Conduct of Research Seminar, Spring 2022

Git Over It: An Introduction to Git for Beginners

Monell Chemical Senses Center, Fall 2021

TEACHING EXPERIENCE

Math and Computer Science Office Hours

Neumann University

November 2025 - Present

- Collaborated with student success team to host office hours and tutor undergraduate students in college algebra, precalculus, discrete math, statistics, calculus, and data mining classes.

Teaching Assistant

August 2024 - May 2025

IS567: Text Mining (Introduction to NLP)

- Collaborated with instructor to grade coursework, answer student's questions, and develop lectures on statistical and transformer-based natural language processing

IS507: Data, Statistical Models, and Information

- Collaborated with instructor and additional TA to develop and grade statistics coursework and answer 80+ students' questions on statistics and statistical modeling

MENTORSHIP AND LEADERSHIP

Seminar Coordinator

August 2024 – August 2025

Amazon-Illinois Center for Interactive Conversational Experiences (AICE) – UIUC-NLP Seminar

- Collaborated with team to schedule and lead weekly seminars featuring speakers from prestigious Universities such as University of Pennsylvania, IBM Research, and Amazon

Mentored Students and Projects

Danni Wu, *Olfactory Information Extraction (Midwest Speech and Language Days, 2026)*, Information Management Masters Student@UIUC 2025-2026

Sameer Komoravolu, *Olfactory Information Extraction (Midwest Speech and Language Days, 2026)*, Computer Science Masters Student@UIUC => Databricks 2026

Jingxuan Wang, *Leveraging Pre-trained Molecular Embeddings and CatBoost for Olfactory Perception Prediction (in progress)*, Data Science Masters Student@University of Michigan 2025

Jainam Rajput, *Scientific Rigor Classification Scoping Review (WCRI 2026)*, Information Management Masters Student@UIUC => Deloitte 2025

PROFESSIONAL SERVICE

Reviewer

AMIA (2026)

KDD AI for Sciences (2026)

Secondary Reviewer

ACL Main Conference (2026)

Journal of Biomedical Informatics (2025)

Panelist

Aspire Illinois Graduate Panel (2025)

Community of Scholars at UIUC Graduate Panel (2025)

Community of Scholars at UIUC Graduate Panel (2024)

PROFESSIONAL ASSOCIATIONS

Association for Computational Linguistics (ACL)

American Chemical Society Division of Chemical Information (ACSCINF)

Association for Information Science and Technology (ASIS&T)

American Medical Informatics Association (AMIA)

Association of Chemoreception Sciences (AChemS)

TECHNICAL SKILLS

- **Programming Languages:** Python, R, Java, SQL, JavaScript, HTML, CSS
- **Machine Learning:** PyTorch, TensorFlow, HuggingFace, scikit-learn, pandas, numpy, matplotlib
- **Technical Skills:** Generative Artificial Intelligence, transformer models, machine learning pipeline, statistics, OOP, OOD, test-driven development, APIs, hypothesis testing, Excel, Git, Tableau